

PROJECT SPECIFICATION

AI Community Skill Exchange Platform

SkillBridge

This document defines the problem, target users, MVP feature scope, technology choices, and data model for the SkillBridge platform. It serves as the primary reference for engineering, design, and product decisions during the build phase.

1. Problem Statement

Skill exchange today is fundamentally broken at the discovery layer. People with valuable expertise — developers, designers, language tutors, electronics repair specialists — have no structured way to surface what they know. Conversely, learners hunting for mentors are stuck with keyword searches across disconnected platforms, no signal on mentor quality for their specific goal, and no roadmap once they find someone.

1.1. Pain Points

For Skill Providers (Mentors)

- No single place to list skills, availability, and experience
- Discovery depends entirely on personal network or word-of-mouth
- No tooling to communicate what they can actually teach vs. what they know

For Learners

- Mentor search is keyword-based, not goal-based
- No AI-generated learning paths to structure self-directed progress
- No match score or compatibility signal to compare mentors
- Progress and completed skills are not portable across platforms

Market Gap

Existing platforms (Codementor, ADPList, LinkedIn Learning) offer either marketplace or content — not a combined, AI-mediated match between a learner's stated goal and the right mentor.

2. Target Users

Learner Persona	Mentor Persona
Riya, 22 — CS graduate Wants to become a Django backend developer but doesn't know where to start or who can guide her. Needs: • A step-by-step learning path • A matched mentor with relevant skills • Feedback on actual projects	Arun, 29 — Senior developer Has 5 years of Django + Flutter experience. Wants to give back, build his reputation, and earn side income. Needs: • Visibility for his skills • Matched learner requests • A profile that sells itself

3. MVP Feature List

#	Feature	Priority
1	Authentication & Roles Email + password, Google OAuth, GitHub OAuth via Supabase Auth. Role selection (Learner / Mentor) at signup. JWT passed to Django for API auth.	Must Have
2	Skill Marketplace Mentors create skill listings (title, category, level, description). Learners browse and filter. Full-text search on title + category.	Must Have
3	AI Skill Matching Learner inputs a free-text goal. Gemini scores all mentors against required skills and returns ranked top-5 with match percentage. Core differentiator.	Must Have — AI Core
4	AI Learning Path Generator Learner submit goals; Gemini outputs a week-by-week roadmap stored as structured JSON. Path is viewable, shareable, and attached to the learner's profile.	Must Have — AI Core
5	Mentorship Request & Basic Chat Learner sends request to mentor. Mentor accepts or rejects. On accept, a Supabase Realtime chat channel is opened for the pair. Read receipts included.	Must Have

4. Tech Stack

Layer	Technology	Version / Notes
Frontend Framework	Next.js	15 — App Router, React Server Components
Frontend Language	TypeScript	Strict mode
Styling	Tailwind CSS + shadcn/ui	Utility-first + accessible component primitives
Backend Framework	Django + DRF	Django 5.x · djangorestframework 3.x
Database	Supabase PostgreSQL	Managed Postgres with RLS policies
Auth	Supabase Auth	Email/Password, Google OAuth, GitHub OAuth
AI / LLM	Gemini API (Google)	gemini for matching + path generation
API Docs	drf-spectacular	Swagger UI at /api/schema/swagger-ui/
Backend Deployment	Railway	Docker-based, auto-deploy from GitHub main
Frontend Deployment	Vercel	Edge network, preview deployments per PR
Realtime / Chat	Supabase Realtime	Postgres Change Data Capture over WebSockets

4.1 · Architecture Overview

All client requests hit the Next.js frontend. API calls are routed to the Django REST API. The Django service communicates with Supabase Postgres via a connection pool (DATABASE_URL). Auth tokens issued by Supabase Auth are validated on Django via the service_role key. Long-running Gemini calls are offloaded to Celery workers backed by Redis to prevent HTTP timeouts.

5. Data Model Sketch

All tables live in Supabase PostgreSQL. UUIDs are used for all primary keys to align with Supabase Auth's uid() function. Row Level Security (RLS) is enabled on all user-owned tables.

user_profiles

Extends Supabase Auth users. One row per registered user.

Column	Type	Constraints	Notes
id	UUID	PK, FK → auth.users	Matches Supabase auth UID
role	VARCHAR(20)	NOT NULL	learner mentor admin
display_name	VARCHAR(100)	NOT NULL	
bio	TEXT	NULLABLE	
avatar_url	TEXT	NULLABLE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

mentor_profiles

Extended detail for mentor-role users. 1-to-1 with user_profiles.

Column	Type	Constraints	Notes
id	UUID	PK, FK → user_profiles	
experience_yrs	SMALLINT	NOT NULL DEFAULT 0	
availability	VARCHAR(50)	NULLABLE	e.g. 5hrs/week
linkedin_url	TEXT	NULLABLE	
ai_summary	TEXT	NULLABLE	AI-generated professional summary

skills

Skill listings created by mentors.

Column	Type	Constraints	Notes
id	UUID	PK DEFAULT gen_random_uuid ()	
mentor_id	UUID	FK → user_profiles(id)	Owning mentor
title	VARCHAR(100)	NOT NULL	

Column	Type	Constraints	Notes
category	VARCHAR(100)	NOT NULL	e.g. Programming
level	VARCHAR(20)	NOT NULL	Beginner Intermediate Advanced
description	TEXT	NOT NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

mentorship_requests

Tracks the lifecycle of a learner's request to a mentor.

Column	Type	Constraints	Notes
id	UUID	PK DEFAULT gen_random_uuid() ()	
learner_id	UUID	<i>FK</i> → <i>user_profiles(id)</i>	
mentor_id	UUID	<i>FK</i> → <i>user_profiles(id)</i>	
status	VARCHAR(20)	NOT NULL DEFAULT pending	pending accepted rejected cancelled
message	TEXT	NULLABLE	Intro message from learner
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

learning_paths

AI-generated week-by-week roadmaps stored as structured JSON.

Column	Type	Constraints	Notes
id	UUID	PK DEFAULT gen_random_uuid() ()	
user_id	UUID	<i>FK</i> → <i>user_profiles(id)</i>	Learner who owns this path
goal	TEXT	NOT NULL	Free-text goal input
title	VARCHAR(200)	NOT NULL	AI-generated title
content	JSONB	NOT NULL	{ weeks: [{ week, topics: [] }] }
created_at	TIMESTAMPTZ	DEFAULT NOW()	

ai_match_results

Cached results of Gemini matching runs. Avoids redundant AI calls for identical goals.

Column	Type	Constraints	Notes
id	UUID	PK DEFAULT gen_random_uuid()	
learner_id	UUID	FK → user_profiles(id)	
goal	TEXT	NOT NULL	
results	JSONB	NOT NULL	[[mentor_id, score, rationale]]
created_at	TIMESTAMPTZ	DEFAULT NOW()	TTL: purge after 24h

5.1 · Relationship Summary

From	Cardinality	To	Via
user_profiles	1 : 0..1	mentor_profiles	mentor_profiles.id
user_profiles	1 : N	skills	skills.mentor_id
user_profiles (learner)	1 : N	mentorship_requests	mentorship_requests.learner_id
user_profiles (mentor)	1 : N	mentorship_requests	mentorship_requests.mentor_id
user_profiles	1 : N	learning_paths	learning_paths.user_id
user_profiles	1 : N	ai_match_results	ai_match_results.learner_id